



CST 438 IMAGE PROCESSING TECHNIQUE 3CR

MODULE 1

Digital Image Fundamentals

An image is a 2D representation of a three dimensional scene. A digital image is basically a numerical representation of an object. The term digital image processing refers to the manipulation of an image by means of a processor. The different elements of an image-processing system include image acquisition, image storage, image processing and display.

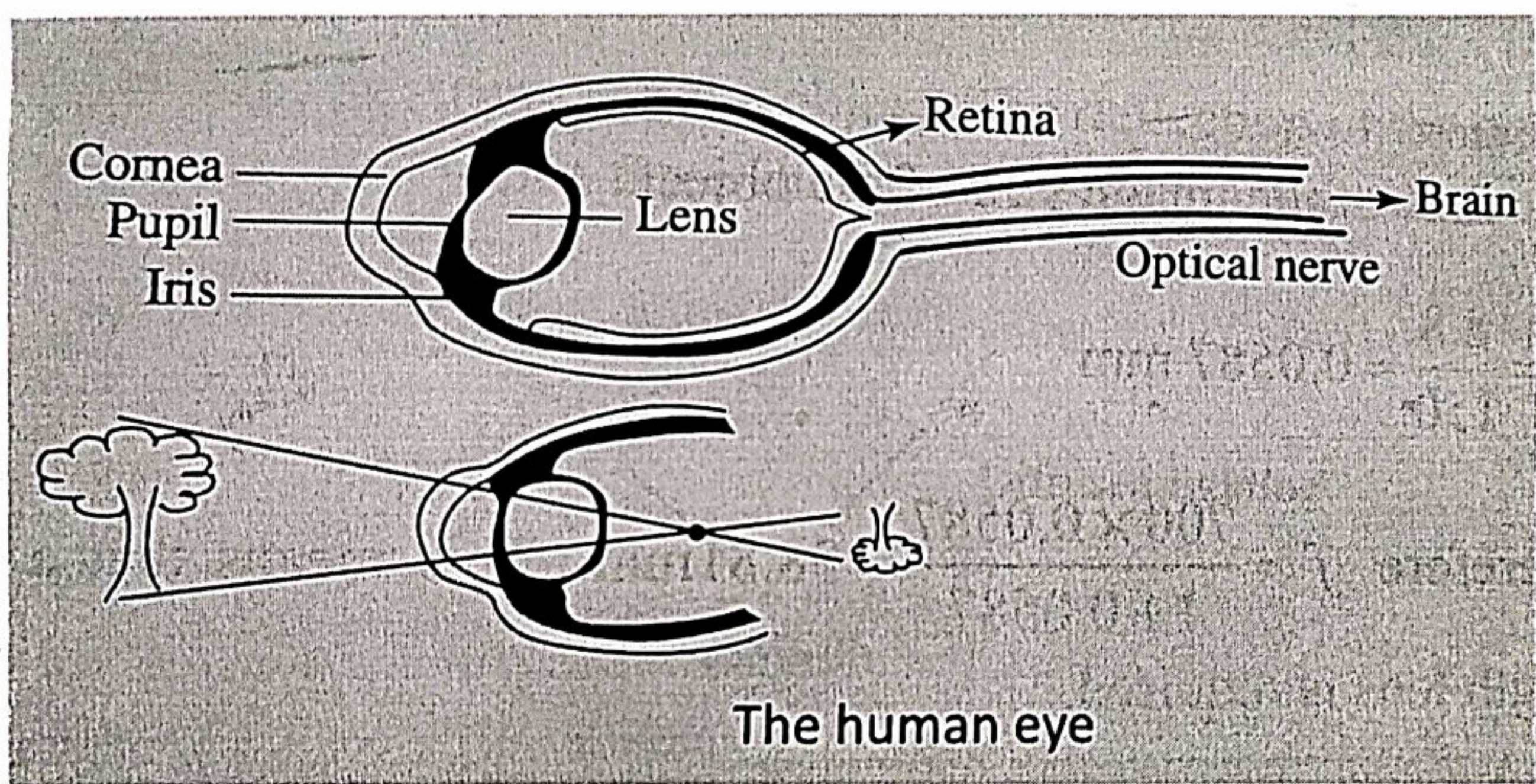
Elements of Visual Perception.

1. Structure of the Human Eye:

The human visual system is made of two components - the human eye and portions of brain associated with the human visual system.

The human eye can be compared to a camera system. The similarities between a camera and the human eye are obvious. Like the camera, the

eye converts the visual information into neural signals. These signals are then carried by the optical nerves to the human brain. The human brain processes these neural signals and produces object perception. The human eye has the components shown in the following figure.



The components of the human eye and their functions are as follows:

1. The cornea is the protective covering of the eye.

This acts like a convex lens refracting the light rays. This is the first level of focusing that the

light entering the eye undergoes. Light then travels through the aqueous and the vitreous humors, which are liquid media that facilitate the travel of light.

2. The iris acts as a variable aperture to control the amount of light that is allowed to pass through the lens. This is controlled by muscles, which open and close based on the average intensity of the scene being viewed.
3. The retina is the eye sensor system and is composed of photoreceptors. The distance between the lens and the retina varies from 14 mm to 17 mm.

The human eye has two kinds of photoreceptors - cones and rods. When the light strikes the cones or rods, it creates electrochemical reactions that generate neural impulses. The neural signals are sent via the optical nerve to the visual cortex region in the brain, where perception is created.



The human eye has three different types of cones. Each cone responds to a distinct spectral band of light. This creates the perception of colour, as the cones respond by generating a unique set of responses for each colour of light. By getting these responses from the three cone types, the brain forms a distinct perception of colour. This is called the concept of trichromacy.

Differences between rods and cones

Rods	Cones
- Number around 100 million - Respond to the broad spectrum of light but are colour blind. - Useful in low light vision - Absent in the region called fovea. - Best view in periphery	- Number around 6-7 million - Perceive colour - Useful in bright light vision - Present more in the region around the fovea. Colour perception is best when the object is directly viewed. - Minimal perception in periphery

2. Properties of Human Visual System

The important properties of the human visual system that are necessary for the study of image processing are as follows:

1. Brightness adaptation

There are two distinct visual systems created by rods and cones. Rods cannot perceive colour, but are sensitive to light. This kind of vision created by the rods is called scotopic vision. Photopic vision is created by the cones as they are sensitive to colour, but are less sensitive to light. There are three types of brightness adaptation - general, local and lateral.

a. General brightness adaptation:

General brightness adaptation refers to the average brightness of the scene.

b. Local brightness adaptation:

The eye quickly adjusts its brightness adaptation before moving to the next object.

c. Lateral brightness adaptation:

Sensitivity changes in the local areas of the object are associated with similar changes in the background.



2. Intensity and brightness.

Brightness is a psycho-visual concept. It is the Perception or sensation of the input light on the brain that produces the concept of brightness.

3. Simultaneous Contrast:

The object appear bright when the background is darker and not so bright when the background is lighter. This is due to the fact that human perception is sensitive to the luminous contrast with respect to the background rather than the absolute luminance values.

4. Mach bands:

Mach band effect is a phenomenon of lateral inhibition of rods and cones, where the sharp intensity changes are attenuated by the visual system.

5. Frequency response:

The shape of the frequency response of the human visual system is similar to that of the band-pass filter.

A Simple Image Formation Model

Images are denoted by two-dimensional functions of the form $f(x, y)$. The value or amplitude of f at spatial coordinate (x, y) is a positive scalar quantity whose physical meaning is determined by the source of the image.

When the image is generated from a physical process its intensity values are proportional to energy radiated by a physical source (e.g.: electromagnetic waves). As a consequence $f(x, y)$ must be non-zero and finite; that is

$$0 < f(x, y) < \infty \dots \dots (1)$$

The function $f(x, y)$ may be characterised by two components:

1. The amount of source illumination incident on the scene being viewed, appropriately called illumination and denoted by $i(x, y)$.
2. The amount of illumination reflected by the objects in the scene, appropriately called reflectance and is denoted by $r(x, y)$.

The two functions combined as a product to form $f(x, y)$:



$$f(x, y) = i(x, y) r(x, y) \dots \dots (2)$$

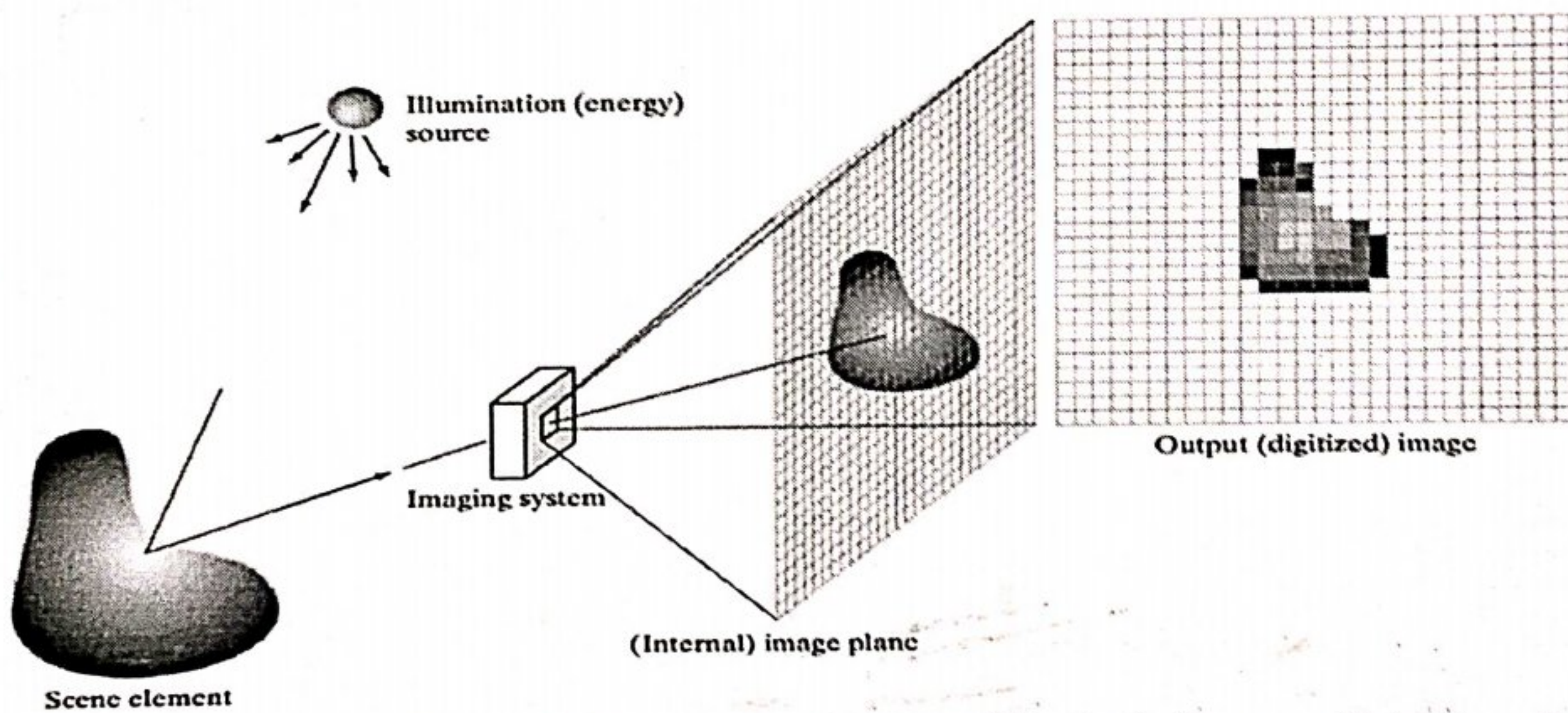
Where

$$0 < i(x, y) < \infty \dots \dots (3)$$

and

$$0 < r(x, y) < 1 \dots \dots (4)$$

Equation (4) indicates that reflectance is bounded by 0 (total absorption) and 1 (total reflectance)



An example of the digital image acquisition process. (a) Energy ("illumination") source. (b) An element of a scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.



Spatial and Intensity Resolution.

Spatial resolution is a measure of the smallest discernible detail in an image. (Intuitively)

Spatial resolution (quantitatively) can be stated in a number of ways, with line pairs per unit distance, and dots (pixels) per unit distance being among the most common measures.

Dots per unit distance is a measure of image resolution used commonly in the printing and publishing industry.

For quality news papers : 75 dpi (dots per inch)

Magazines : 133 dpi

glossy brochure : 175 dpi

Text books : 2400 dpi

Intensity resolution refers to the smallest discernible change in intensity level.

Image Interpolation

Interpolation is a basic tool used extensively in tasks such as zooming, shrinking, rotating and geometric operations.

Fundamentally, interpolation is the process of using known data to estimate values at unknown locations.



Eg:- Suppose that an image of size 500×500 pixels has to be enlarged 1.5 times to 750×750 pixels.

1. Nearest neighbor Interpolation

A simple way to visualize zooming is to create an imaginary 750×750 grid with the same pixel spacing as the original, and then shrink it so that it fits exactly over the original image. Obviously, the pixel spacing in the shrunken 750×750 grid will be less than the pixel spacing in the original image. This method is called nearest neighbor interpolation. In this each new location is assigned by the intensity of its nearest neighbor in the original image. This approach is simple but it has the tendency to produce undesirable artifacts, such as severe of straight edges.

2. Bilinear Interpolation:

A more suitable approach, which uses the four nearest neighbors to estimate the intensity at a given location. Let (x, y) denote the coordinates of the location to which we want to assign an intensity value, and let $v(x, y)$ denote that intensity value. For bilinear interpolation, the assigned value is obtained using the equation

$$v(x, y) = ax + by + cxy + d$$

Where the four coefficients are determined from the four equations in four unknowns that can be written using the four nearest neighbors of point (x, y) .

3. Bicubic interpolation, involves the sixteen nearest neighbors of a point. The intensity value assigned to point (x, y) is obtained using the equation:

$$V(x, y) = \sum_{i=0}^3 \sum_{j=0}^3 a_{ij} x^i y^j$$

Classification of Digital Images

Digital images can be broadly classified into two types and they are (i) raster image, and (ii) vector image.

1. Raster Image or Bitmap Image

A raster image file is generally defined as a rectangular array of regularly sampled values known as pixels. A raster image is resolution dependent because it contains a fixed number of pixels that are used to create the image.

Common raster image formats include BMP (Windows Bitmap), PCX (Paintbrush), TIFF (Tag Interleave Format), JPEG (Joint Photographics Expert Group), GIF (Graphics Interchange Format), PNG (Portable Network Graphics), PSD (Adobe Photoshop), and CPT (Corel

PhotoPaint)

2. Vector Image

A vector image is defined by objects which are made of lines and curves that are mathematically defined in the computer. A vector can have various attributes such as line thickness, length and colour. Vector images are suitable for typography, line art and illustrations.

IMAGE TYPES

Images can be broadly classified under four categories:

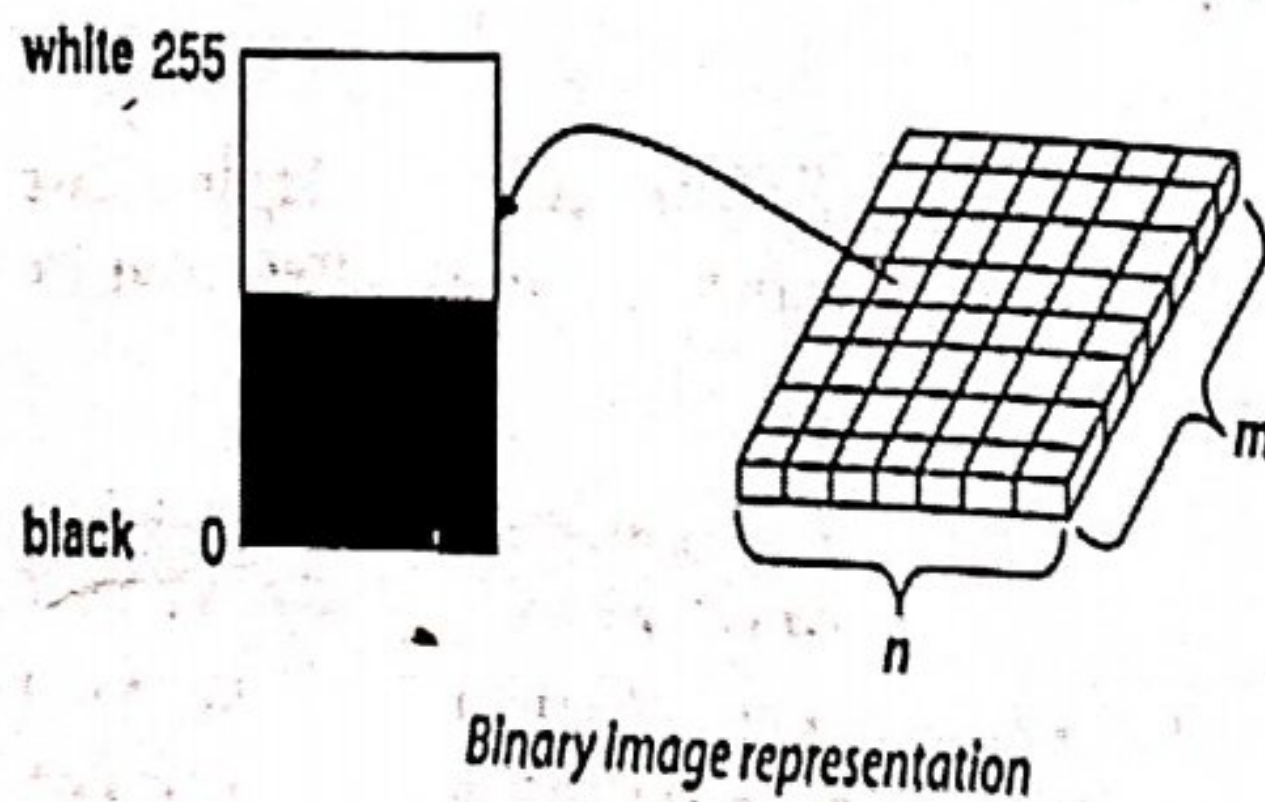
- 1) Black and white or binary images
- 2) Grayscale images.
- 3) Colour images.
- 4) Multispectral images.

1. Binary Images.

Binary images take only two values, i.e., either '0' or '1'. The brightness gradation cannot be differentiated in the binary image.

2. Grayscale Images

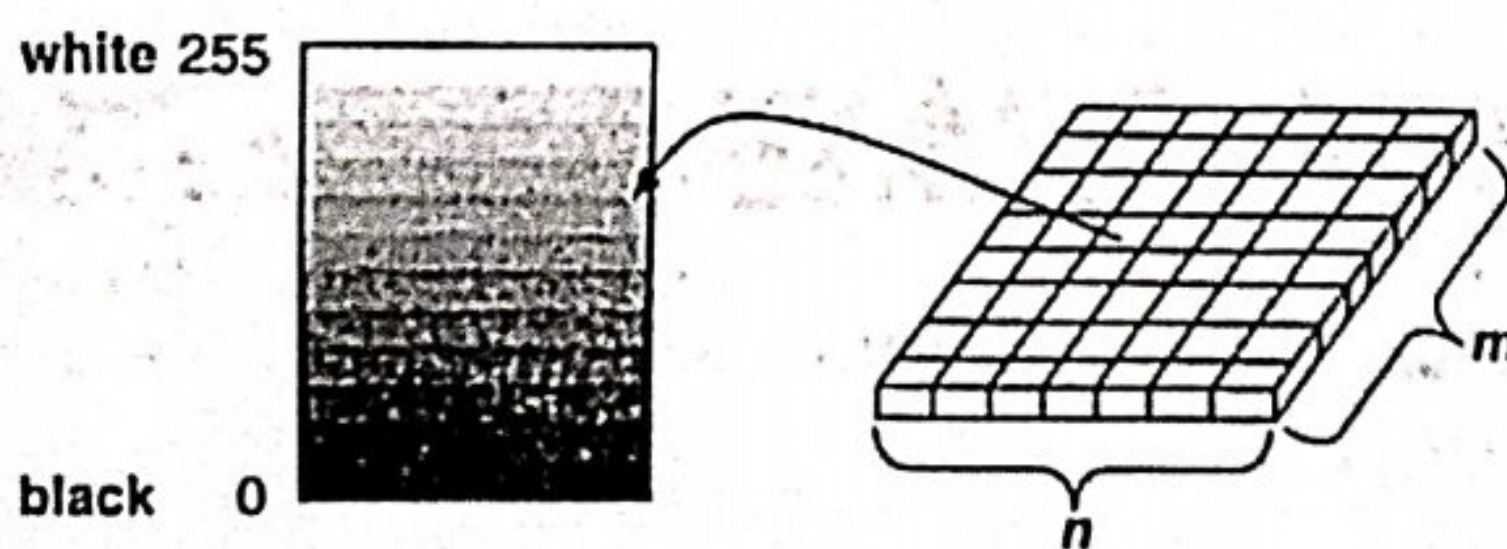
Grayscale images contain only brightness information. Each pixel value in a grayscale image corresponds to an amount or quantity of light. The brightness graduation can be differentiated in a grayscale image. An 8-bit image will have a brightness variation from 0 to 255 where '0' represents black and '255' represents white. A grayscale image measures only the light intensity.



Binary Image representation



Mother Teresa in black and white



Grayscale Image representation



Mother Teresa in grayscale

3. Colour Image

A colour image has three values per pixel and they measure the intensity and chrominance of light. Each pixel is a vector of colour components. Colour images can be modeled as three-band monochrome image data, where each band of data corresponds to a different colour.

Common colour spaces are RGB (Red, Green, Blue), HSV (Hue, Saturation, Value) and CMYK (Cyan, Magenta, Yellow, Black).

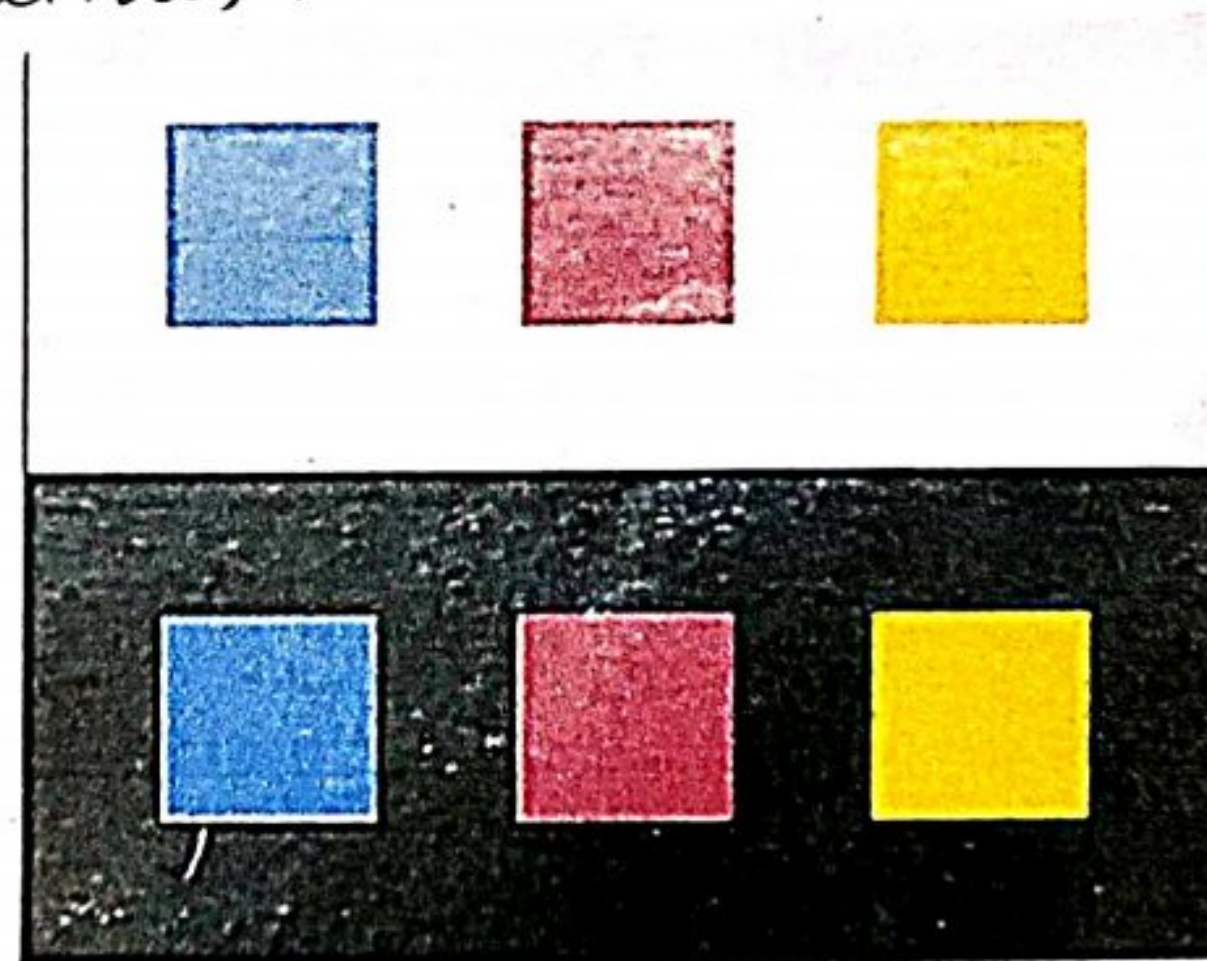


Fig. 1.19 Simultaneous contrast in a colour image

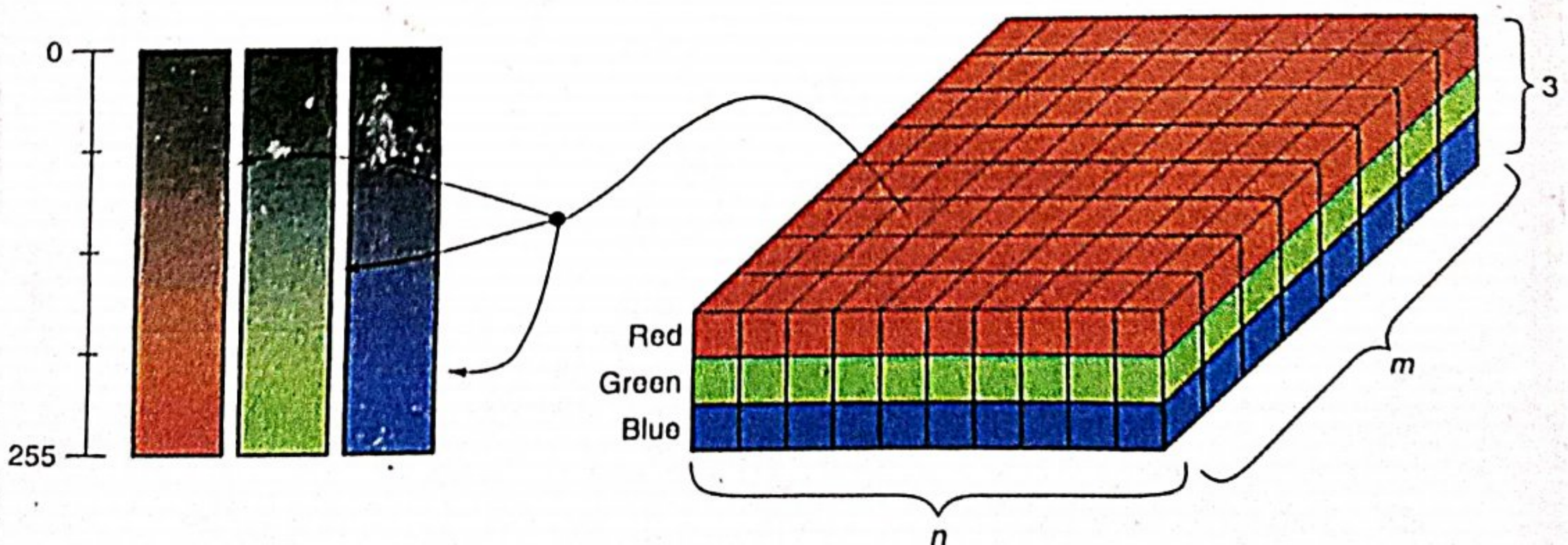


Fig. 1.27 RGB image representation

4. Volume Image

A 3D image is an example of volume image. The volume image can be obtained from some medical imaging equipment in which the individual data points are called 'voxels'. Voxels stands for volume pixels.

Eg:- CAT scan image.

5. Range Image (Depth Image)

Range images are a special class of digital images. Each pixel of a range image expresses the distance between a known reference frame and a visible point in the scene. The range image reproduces the 3D structure of a scene.

6. Multispectral Image

Multispectral images typically contain information outside the normal human perceptual range (infrared, ultraviolet and other bands in electromagnetic spectrum). The information available in a multispectral image is not visible to a human observer.

IMAGE FILE FORMATS

A digital image is often encoded in the form of binary files for the purpose of storage and transmission. A file format is a method used to store digital data and different file formats exist for storing images. Different file formats may compress the image data by differing amounts.

The image file consists of two parts:

- 1) File Header
- 2) Image Data

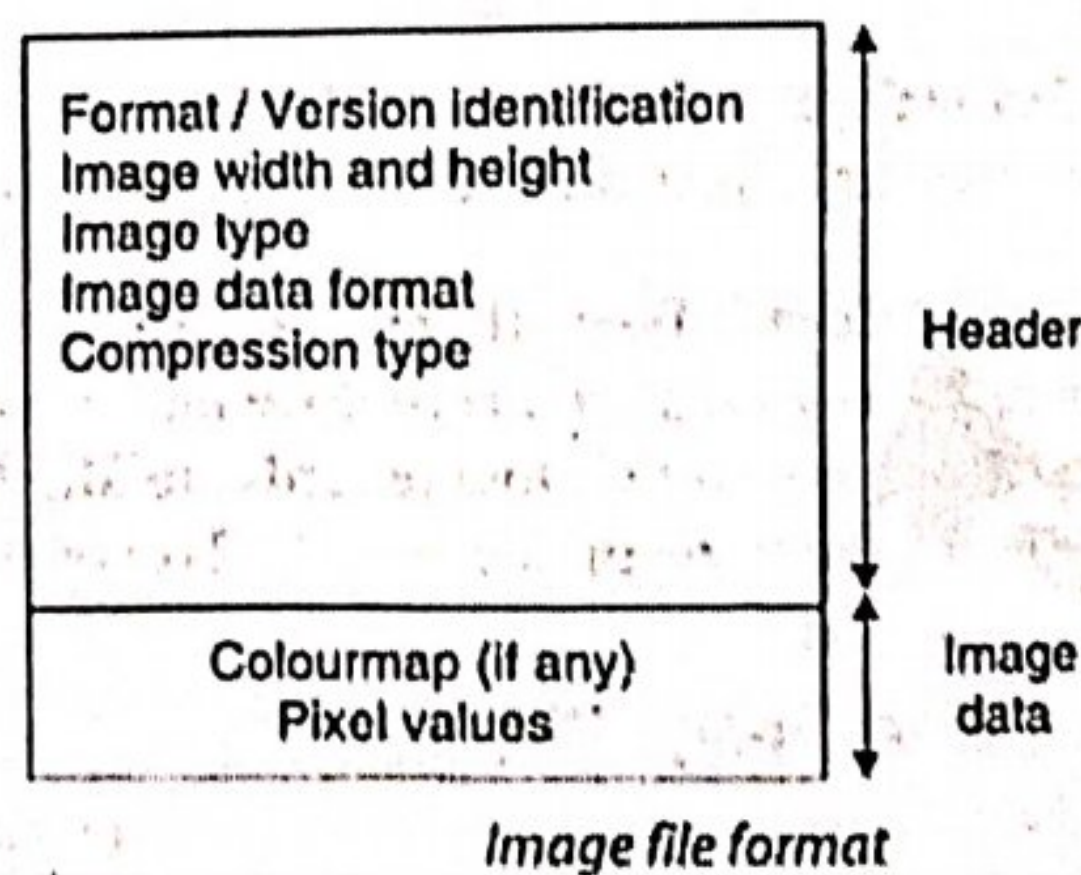
Header Information

The header part contains some of the vital information like format or version identification, width and height of the image, type of the image (binary, grayscale, colour image), and image data format which specifies the order in which pixel values are stored in the image data section. The header also specifies the type of compression mechanism. The length of the file header is often fixed.

Common Image File Formats.

Various image file formats are widely used for the purpose of digital image storage and retrieval. The common image file formats are:

1. GIF (Graphics Interchange Format)
2. JPEG (Joint Photographic Experts Group).
3. PNG (Portable Network Graphics)
4. TIFF (Tagged Image File Format).
5. PSD file format.
6. EPS (Encapsulated PostScript).



Arithmetic and Logical Operations

Arithmetic Operations

- Arithmetic operations are carried out between corresponding pixel pairs.
- Arithmetic operations between images are array operations.
- The four arithmetic operations are denoted as:

$$s(x, y) = f(x, y) + g(x, y)$$

$$d(x, y) = f(x, y) - g(x, y)$$

$$p(x, y) = f(x, y) \times g(x, y)$$

$$v(x, y) = f(x, y) \div g(x, y).$$

The operations are performed between the corresponding pixel pairs in f and g for $x = 0, 1, 2, \dots, M-1$ and $y = 0, 1, 2, \dots, N-1$ where, M and N are the row and column sizes of the images.

- Also, s, d, p & v are images of size $M \times N$.
- Image arithmetic can be performed between images of the same size ($M \times N$).

Applications of Arithmetic operations in Image Processing.

- Addition can be used to form a corrupted image by adding some noise to the noiseless image.

$$g(x, y) = f(x, y) + n(x, y).$$

$\uparrow$ $\uparrow$ $\uparrow$
 Corrupted noiseless Noise.
 Image Image

- A frequent application of image subtraction is in the enhancement of differences between images.

$$g(x, y) = f(x, y) - h(x, y) \text{ (a mask)}$$

- Application of image multiplication and division is shading correction.

$$g(x, y) = f(x, y) \cdot h(x, y)$$

$\uparrow$ $\uparrow$ $\uparrow$
 Output Perfect Shading
 image image function

- Another common use of image multiplication is in masking, also called region of interest (ROI) operations.

Logical Operations

Basic Set operations.

Let A be a set composed of ordered pairs of real numbers. If $a = (a_1, a_2)$ is an element of A , then we write,

$$a \in A$$

Similarly if a is not an element of A , we write

$$a \notin A.$$

The set with no elements is called the null or empty set and is denoted by the symbol $\emptyset$.

$A \subseteq B$; A is said to be a subset of B .

$C = A \cup B$; The union of two sets A and B . (Either A or B)

$D = A \cap B$; The intersection of two sets A and B (Both A and B)

$A \cap B = \emptyset$; A & B are said to be mutually exclusive or disjoint if they have no common elements.

$A^c = \{w | w \notin A\}$; The complement of a set A is the set of elements that are not in A .

$A - B = \{w | w \in A, w \notin B\} = A \cap B^c$; Difference of two sets.

Logical Operations

In case of binary image representation we need to deal with foreground (1-valued) and background (0-valued) set of pixels.

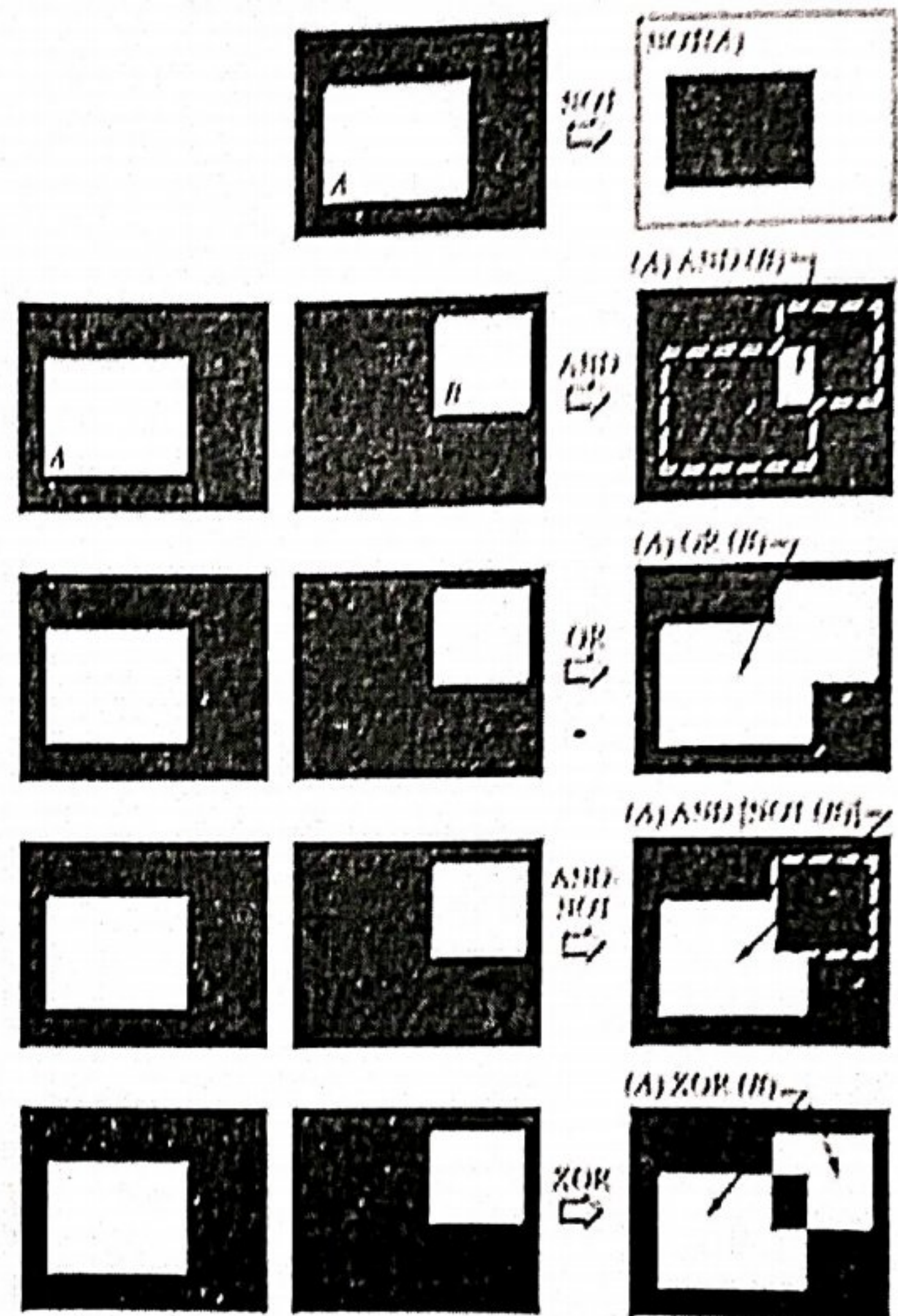
When dealing with binary images, it is common practice to refer to union, intersection and complement as the OR, AND, and NOT logical operations, where "logical" arises from logic theory in which 1 and 0 denote true and false respectively.

Consider two regions (sets) A and B composed of foreground pixels. The OR of these two sets is the set of elements (coordinates) belonging either to A or B or to both. The AND operation is the set of elements that are common to A and B. The NOT operation of a set A is the set of elements not in A. XOR (exclusive OR) operation works on set of foreground pixels belonging to A or B but not both.

→ Illustration of logical operations involving foreground (white) pixels. Black represents binary 0s and white binary 1s. The dashed lines are shown for reference only. They are not part of the result.



Illustration of logical operations involving foreground (white) pixels. Black represents binary 0s and white binary 1s. The dashed lines are shown for reference only. They are not part of the result.



Geometric Spatial Transformations & Image Registration

Geometric Spatial Transformations:

Geometric transformations modify the spatial relationship between pixels in an image. These transformations often are called rubber-sheet transformations because they may be viewed as analogous to "printing" an image on a sheet of rubber and then stretching the sheet according to a predefined set of rules.

A geometric transformation consists of two basic operations:

- 1) Spatial transformation of coordinates and
- 2) Intensity interpolation.

The transformation of coordinates may be expressed as.

$$(x, y) = T \{ (v, w) \} \quad \text{--- (1)}$$

Where (v, w) are pixel coordinates in the original image and (x, y) are the corresponding pixel coordinates in the transformed image.

Example:

The transformation $(x, y) = T \{ (v, w) \} = (v/2, w/2)$ shrinks the original image to half its size in both spatial directions.

$$[x \ y \ 1] = [v \ w \ 1] T = [v \ w \ 1] \begin{bmatrix} t_{11} & t_{12} & 0 \\ t_{21} & t_{22} & 0 \\ t_{31} & t_{32} & 1 \end{bmatrix} \quad (2)$$

This transformation can scale, rotate, translate or sheer a set of coordinate points, depending on the value chosen for the elements of matrix T .

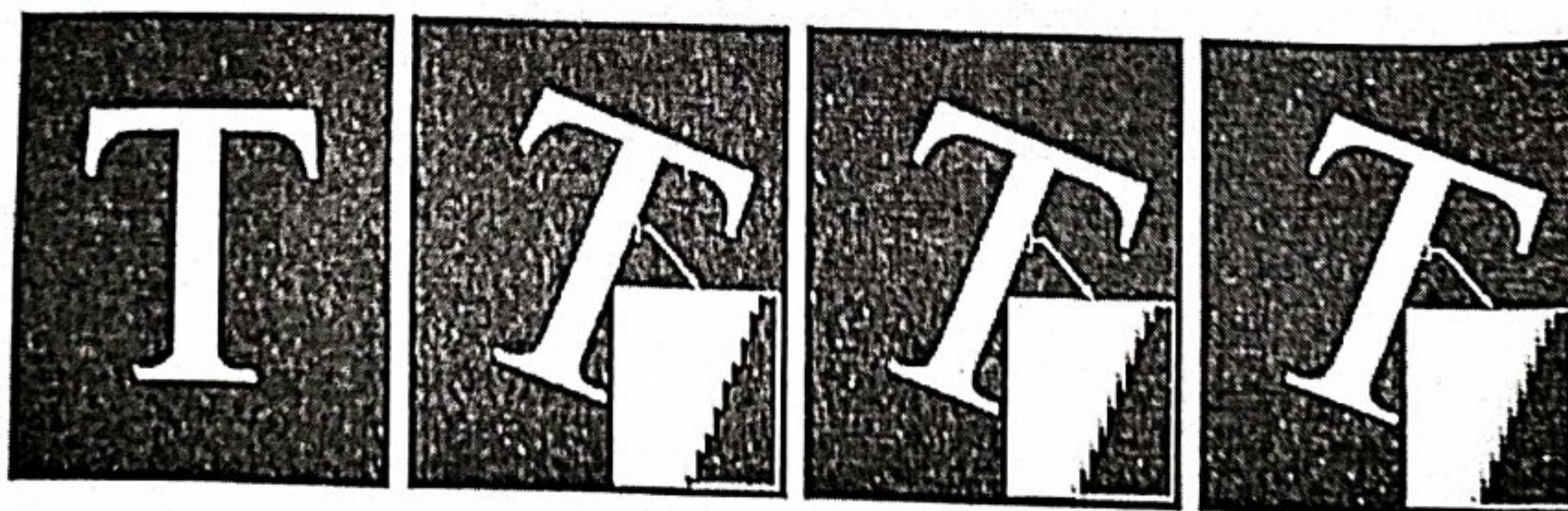
In practice, we can use eqn (2) in two basic ways.

1. Forward mapping: Consists of scanning the pixels of the input image and, at each location (v, w) , computing the spatial location (x, y) , of the corresponding pixel in the o/p image using equation (2).

2. Inverse Mapping: Scans the output pixel locations and at, each location (x, y) computes the corresponding location in the input image using $(v, w) = T^{-1}(x, y)$.

Affine transformations

Transformation Name	Affine Matrix, T	Coordinate Equations	Example
Identity	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = v$ $y = w$	
Scaling	$\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = c_x v$ $y = c_y w$	
Rotation	$\begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = v \cos \theta - w \sin \theta$ $y = v \sin \theta + w \cos \theta$	
Translation	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ t_x & t_y & 1 \end{bmatrix}$	$x = v + t_x$ $y = w + t_y$	
Shear (vertical)	$\begin{bmatrix} 1 & 0 & 0 \\ s_v & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = v + s_v w$ $y = w$	
Shear (horizontal)	$\begin{bmatrix} 1 & s_h & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = v$ $y = s_h v + w$	



a b c d

(a) A 300 dpi image of the letter 'T'. (b) Image rotated 21° using nearest neighbor interpolation to assign intensity values to the spatially transformed pixels. (c) Image rotated 21° using bilinear interpolation. (d) Image rotated 21° using bicubic interpolation. The enlarged sections show edge detail for the three interpolation approaches.

Image Registration

Image registration is an important application of digital image processing used to align two or more images of the same scene. In image registration, we have available the input and output images, but the specific transformation that produced the output image from the input generally is unknown.

- The input image is the image that we wish to transform, and what we call the reference image is the image against which we want to register the i/p.
- One of the principal approaches for solving the problem of geometric distortions caused by differences in viewing angle, distance, and orientation etc., is to use tie points (also called control points), which are corresponding points whose locations are known precisely in the input and reference images.

Input image, control tie point

A simple model based on a bilinear approximation is given by.

$$x = C_1 v + C_2 w + C_3 vw + C_4 \quad \text{--- (3)}$$

$$y = C_5 v + C_6 w + C_7 vw + C_8 \quad \text{--- (4)}$$

Where, during the estimation phase, (v, w) and (x, y) are the coordinates of tie points in the input and reference images, respectively.

If we have four pairs of corresponding tie points in both images, we can write eight equations using Eqns (3) & (4) and use them to solve for the eight unknown coefficients $C_1, C_2, \dots, C_8$. These coefficients constitute the model that transforms the pixels of one image into the locations of the pixels of the other to achieve registration.

Eqns (3) and (4) become the vehicle for transforming all the pixels in the input image to generate the desired new image, which, if the tie points were selected correctly, should be registered with the reference image.



Colour Fundamentals and Colour Models

Colour Fundamentals.

Colour image processing is divided into two major areas: full colour and pseudocolour processing.

* In the first category, the images in question are typically acquired with a full-colour sensor such as a colour TV camera or a colour scanner.

* In the second category, the problem is one of assigning a colour to a particular monochrome intensity or range of intensities.

- The colour spectrum may be divided into six broad regions: violet, blue, green, yellow, orange and red.

Characterization of light is central to the science of color. If the light is achromatic (void of colour), its only attribute is intensity or amount.

- Achromatic light is what viewer seen on a black and white television set.

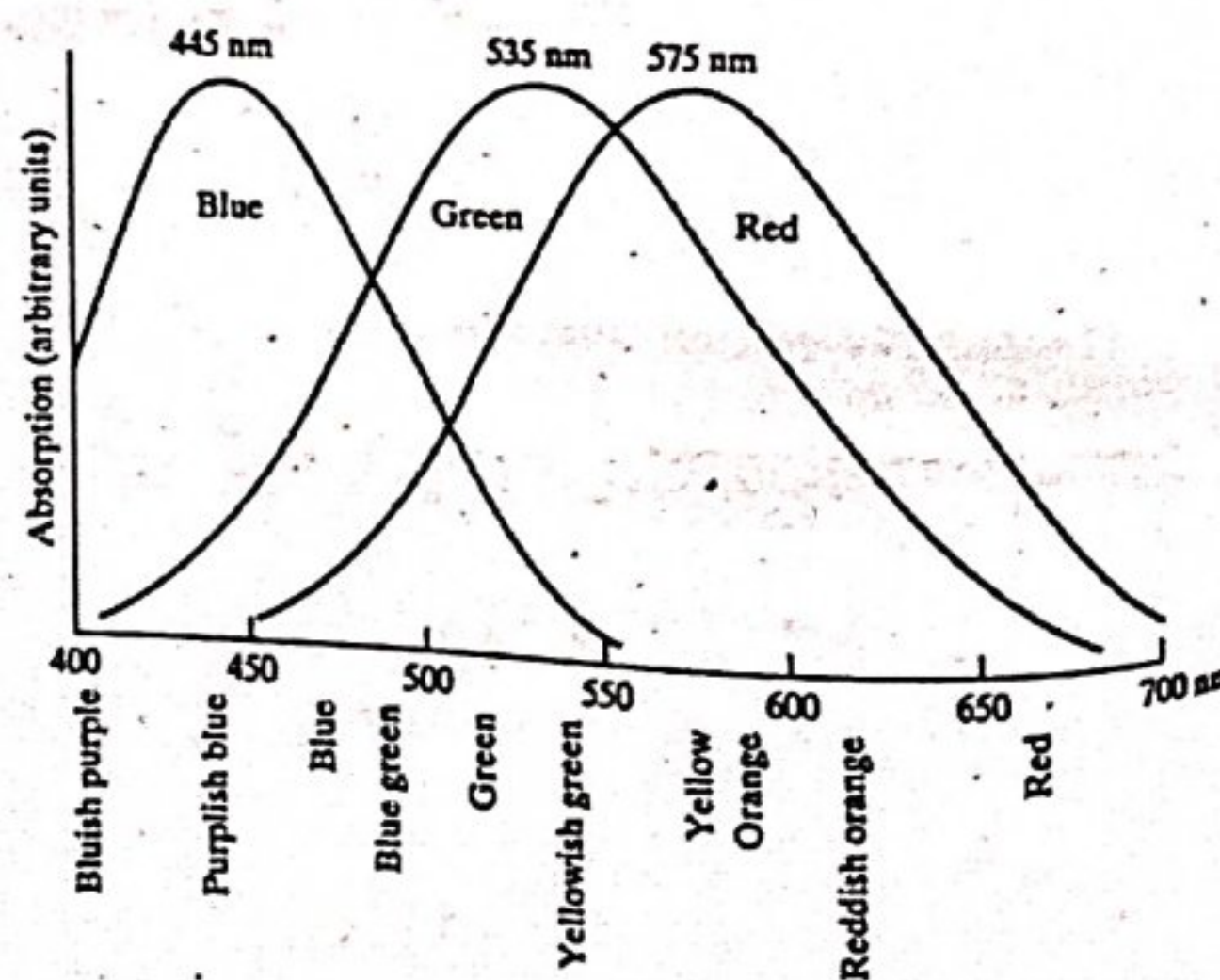
Chromatic light spans the electromagnetic spectrum from approximately 400 to 700 nm. Three basic quantities are used to describe the quality of a chromatic light source: radiance, luminance, and brightness.

Radiance is the total amount of energy that flows from the light source, and is usually measured in watts (W).

Luminance measured in lumens (lm) gives a measure of the amount of energy an observer perceives from a light source.

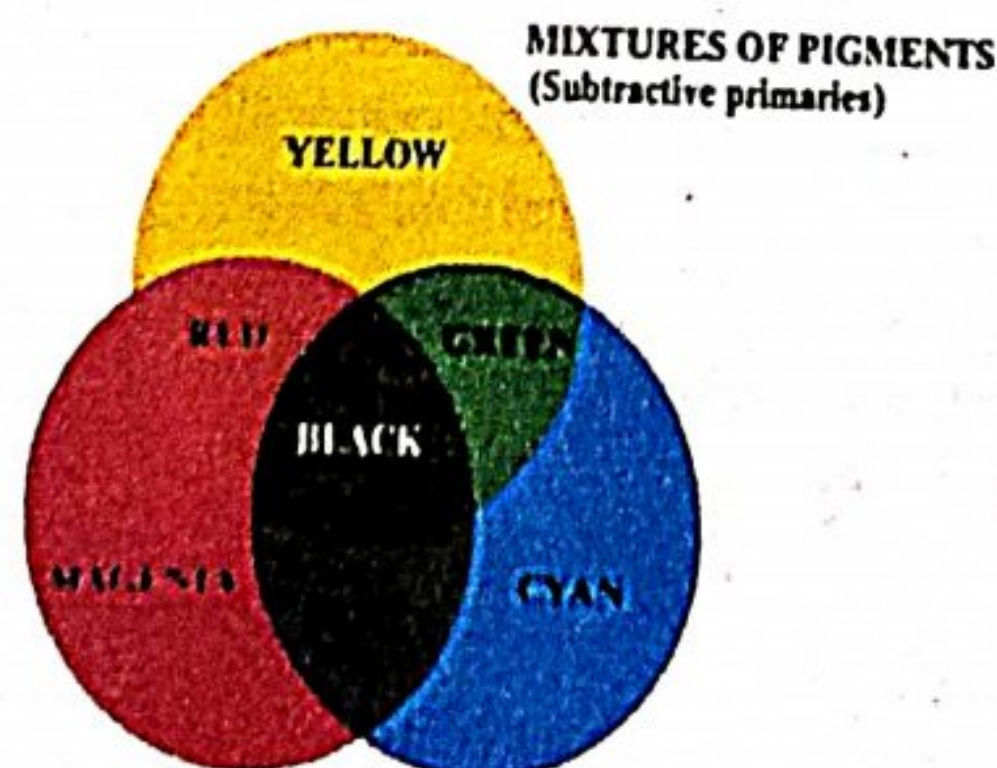
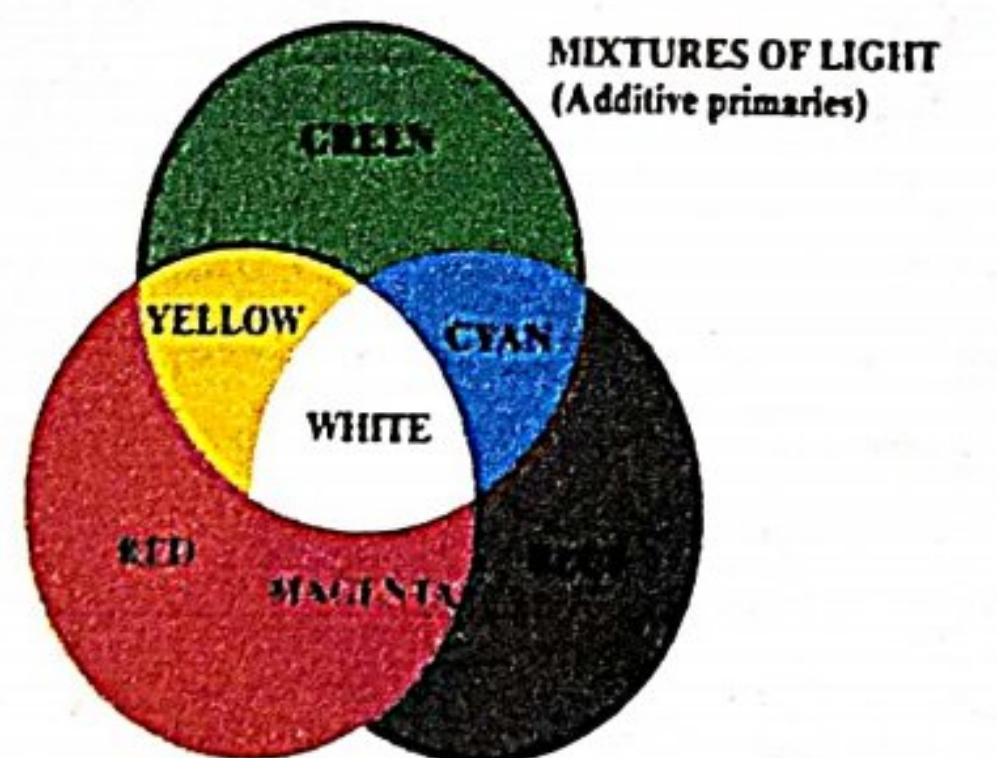
Brightness is a subjective descriptor that is practically impossible to measure.

Absorption of light by the red, green, and blue cones in the human eye as a function of wavelength.



Due to the absorption characteristics of the human eye, colors are seen as variable combinations of the so-called Primary colors red (R), green (G), and blue (B). For the purpose of standardization, the CIE (International Commission on Illumination) designated in 1931 the following specific wavelength values to the three primary colours: blue = 435.8 nm, green = 546.1 nm and red = 700 nm.

The primary colours can be added to produce the secondary colours of light - magenta (red + blue), cyan (green + blue) and yellow (red + green).



PRIMARY AND SECONDARY COLORS
OF LIGHT AND PIGMENT

FIGURE 6.4
Primary and secondary colors of light and pigments. (Based on General Electric Co., Lamp Business Division.)

The characteristics generally used to distinguish one colour from another are brightness, hue, and saturation.

Brightness embodies the achromatic notion of intensity.

Hue is an attribute associated with the dominant wavelength mixture of light waves. Hue represents dominant colour as perceived by an observer.

Saturation refers to the relative purity or the amount of white light mixed with a hue.

- Hue & saturation taken together are called chromaticity.

- A colour may be characterized by its brightness and chromaticity.

The amount of red, green, and blue needed to form any particular colour are called the tristimulus values and are denoted x , y , and z respectively. A colour is specified by its trichromatic coefficients defined as

$$x = \frac{x}{x + y + z} \quad \text{---(1)}$$

$$y = \frac{y}{x + y + z} \quad \text{---(2)}$$

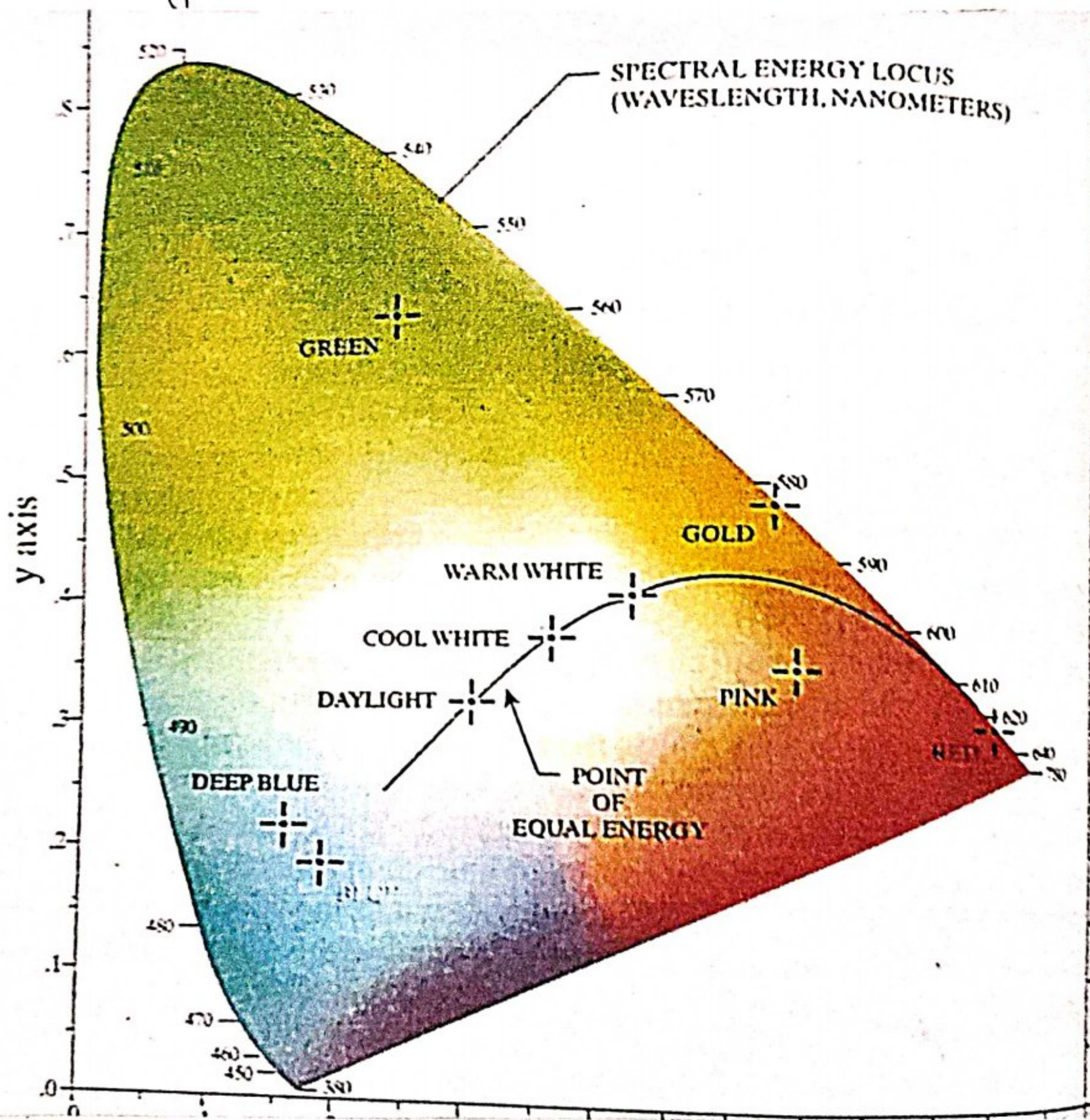
$$z = \frac{z}{x + y + z} \quad \text{---(3)}$$

$$x + y + z = 1 \quad \text{--- (4)}$$

For any wavelength of light in the visible spectrum, the tristimulus values needed to produce the color corresponding to that wavelength can be obtained directly from curves or tables that have been compiled from extensive experimental results.

Another approach for specifying colors is to use the CIE chromaticity diagram, which shows color composition as a function of x (red) and y (green).

$$z = 1 - (x + y).$$



Colour Models

The purpose of a colour model (also called colour space or colour system) is to facilitate the specification of colours in some standard, generally accepted way.

A colour model is a specification of a coordinate system and a subspace within that system where each colour is represented by a single point.

The hardware oriented models most commonly used in practice are the RGB (Red, Green, Blue) model for colour monitors and a broad class of colour video cameras; the CMY (Cyan, Magenta, Yellow) and CMYK (Cyan, Magenta, Yellow, Black) models for colour printing and the HSI (hue, saturation, intensity) model, which corresponds closely with the way human describes and interpret colour.

1. RGB Colour Model

In an RGB colour model, the three primary colours red, green and blue form the axis of the cube which is illustrated in the following figure. Each point in the cube represents a specific colour.



This model is good for setting the electron gun for a CRT.

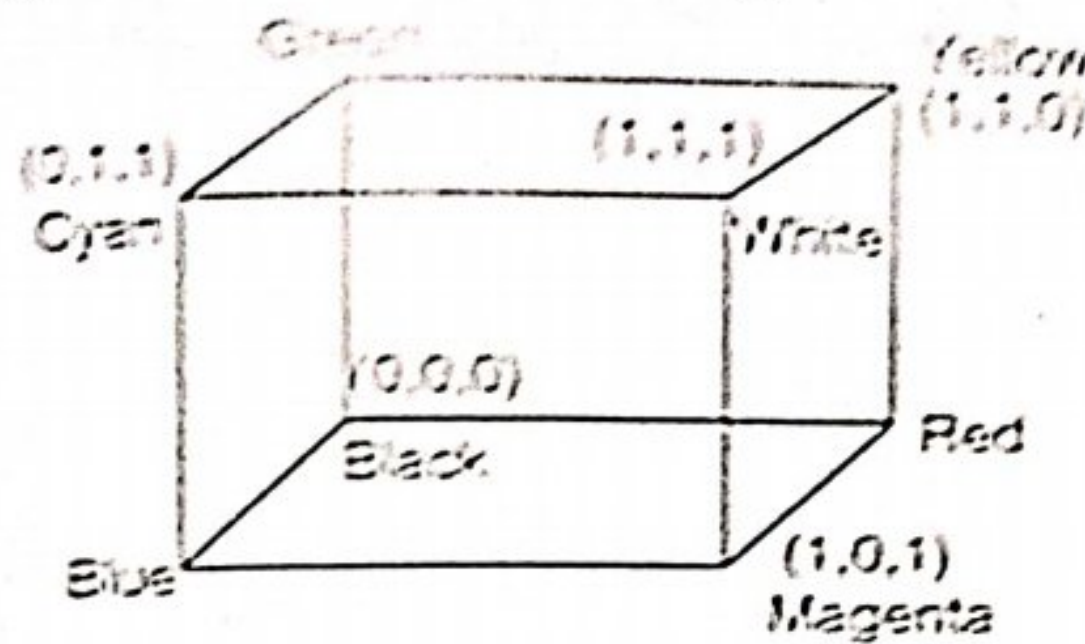


Fig.11.2 RGB colour cube

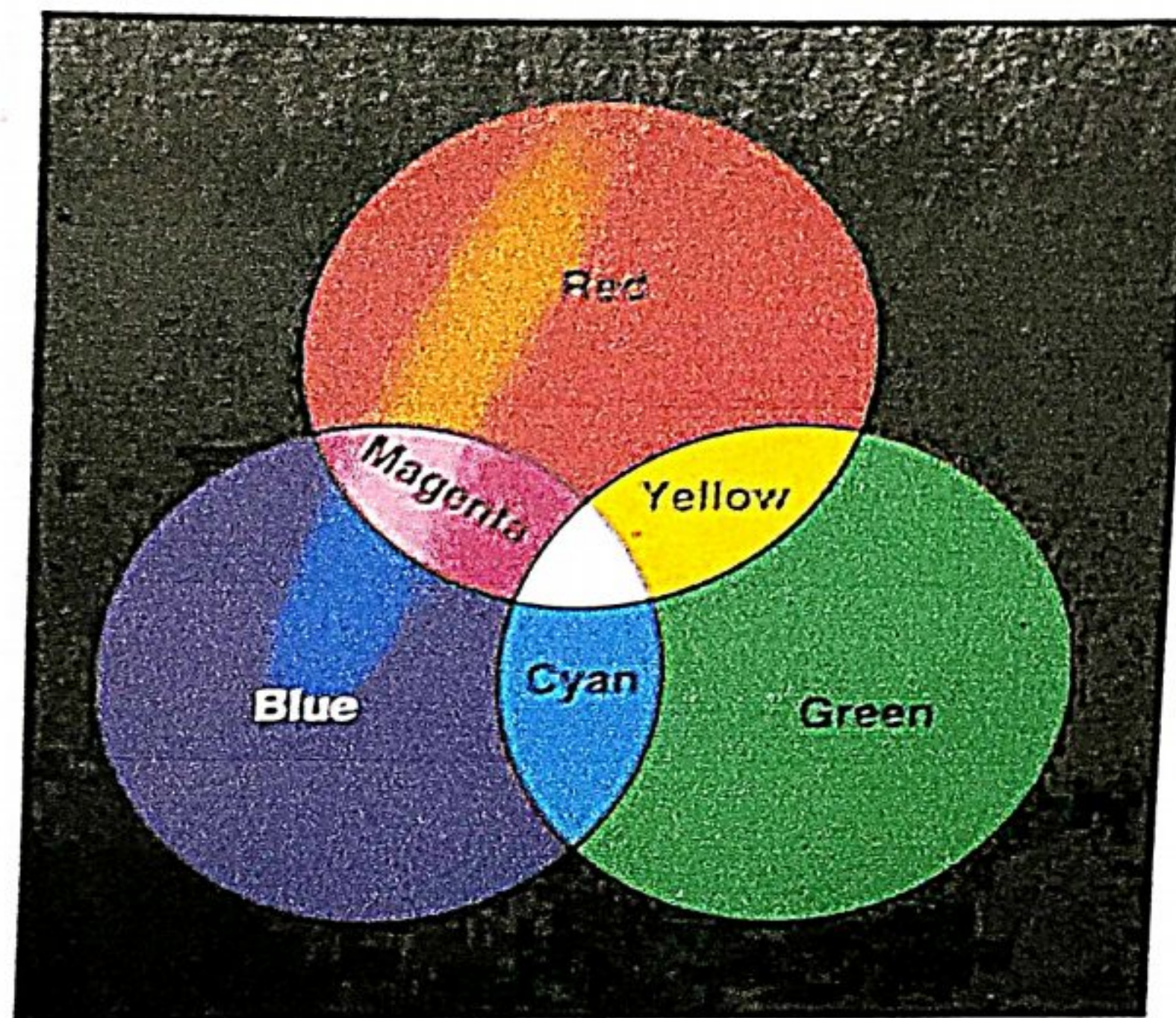


Fig.11.3 RGB colour model

RGB is an additive colour model, From the above figure it is obvious that

$$\text{Magenta} = \text{Red} + \text{Blue}$$

$$\text{Yellow} = \text{Red} + \text{Green}$$

$$\text{Cyan} = \text{Blue} + \text{Green}$$

Limitations of the RGB Model:

1. The RGB colour coordinates are device dependent. This implies that the RGB model will not in general reproduce the same colour from one display to another.
2. The RGB model is not perceptually uniform.
3. It is difficult to relate this model to colour appearance because its basis is to device signals and not display luminance values.

2. CMY Colour Model.

Printers commonly employ the CMY model. Printers produce an image by reflective light, which is basically a subtractive process. The conversion between RGB and CMY is done as follows:

$$C = 1 - R$$

$$M = 1 - G$$

$$Y = 1 - B$$

CMY is a subtractive colour model.

$$\text{Magenta} = \text{White} - \text{Green.}$$

$$\text{Cyan} = \text{White} - \text{Red}$$

$$\text{Yellow} = \text{White} - \text{Blue}$$

The CMY colour cube and model are shown in the following figures.

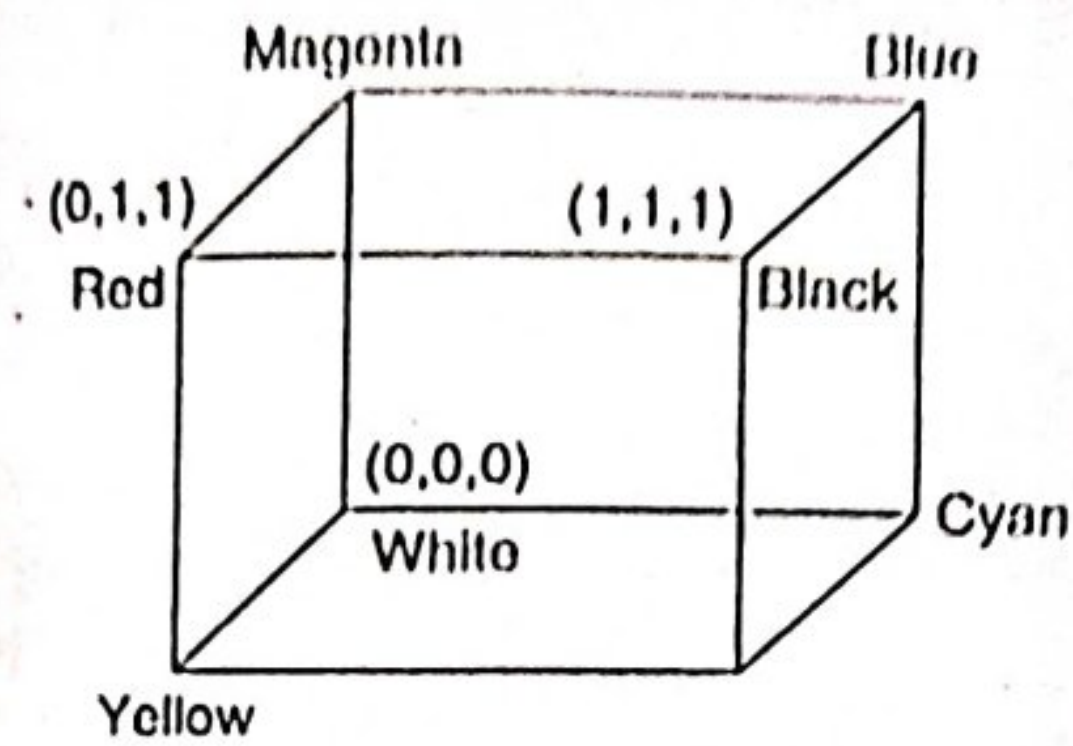


Fig. 11.6 CMY colour cube

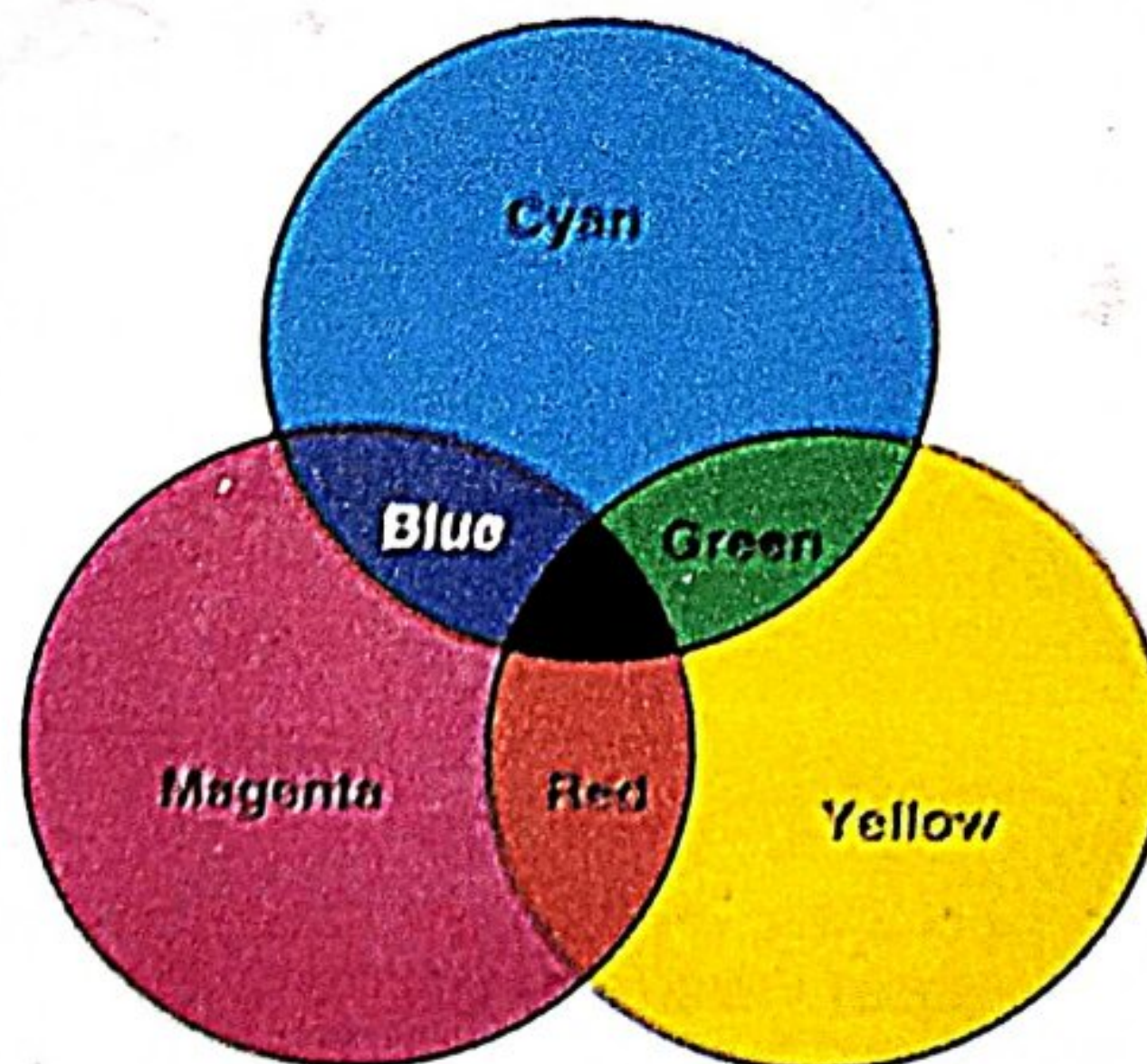


Fig. 11.7 CMY colour model

3. HSI Colour Model

HSI stands for Hue, Saturation and Intensity.

Hue represents dominant colours as perceived by an observer. It is an attribute associated with the dominant wavelength. Saturation refers to the relative purity or the amount of white light mixed with a hue.

Intensity reflects the brightness.

The conversion from RGB space to HSI space is given below.

$$I = \frac{1}{3} (R + G + B)$$

$$S = 1 - \frac{3}{(R + G + B)} [\min(R, G, B)]$$

$$H = \cos^{-1} \left\{ \frac{0.5 [(R - G) + (R - B)]}{\sqrt{(R - G)^2 + (R - B)(G - B)}} \right\}$$

The HIS/HSI colour model can be considered as cylinder, where the coordinates, r , θ , and z are saturation, hue and intensity respectively.

The cylindrical representation of the HSI colour model is illustrated in the following figure.

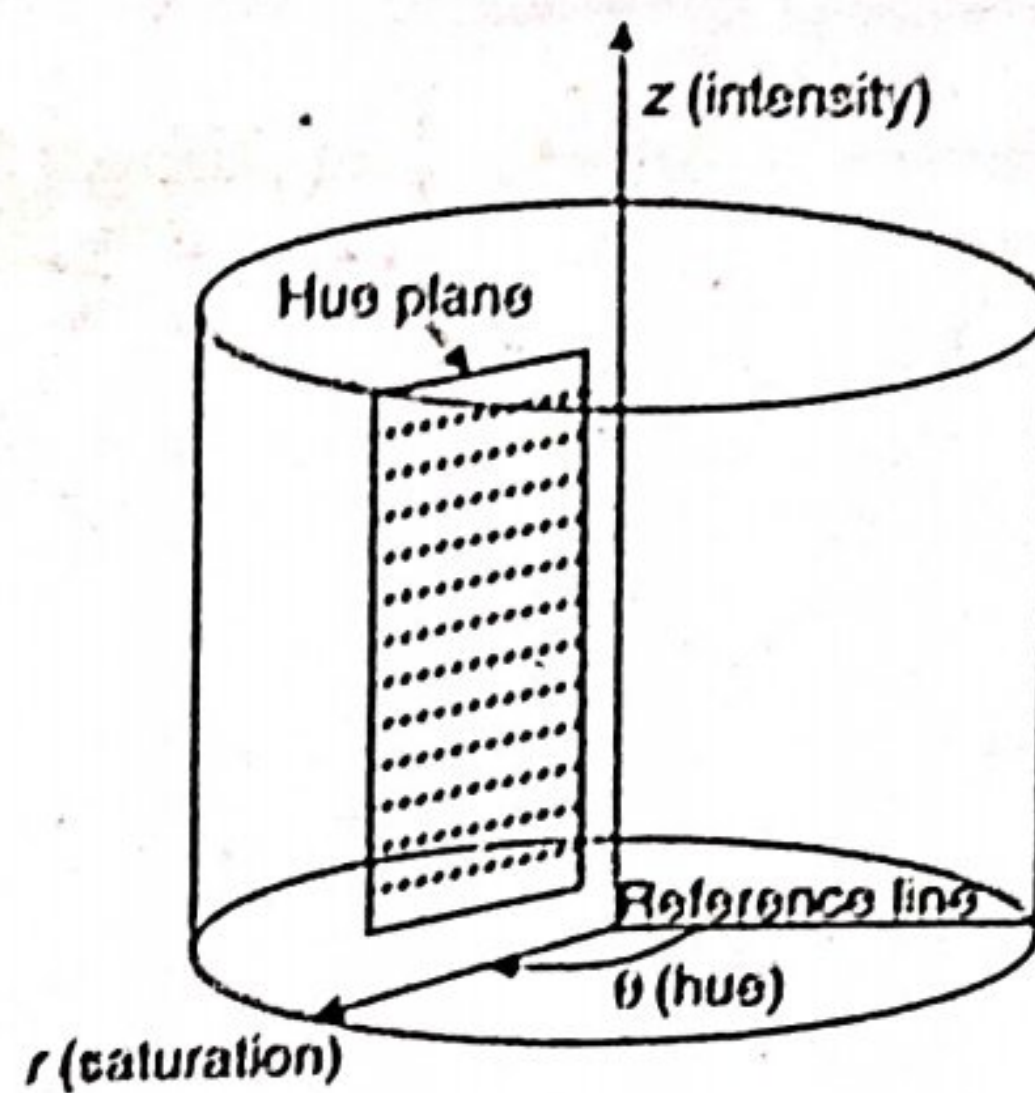


Fig.11.8 Cylindrical representation of HIS colour model

4. YIQ Colour Model:

The YIQ colour model is defined by the National Television System Committee (NTSC). In this model, Y represents the luminance; and I and Q describe the chrominance. Conversion between RGB and YIQ is given below:

$$\begin{bmatrix} Y \\ I \\ Q \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ 0.596 & -0.274 & -0.322 \\ 0.211 & -0.523 & 0.311 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix} \quad \text{and}$$

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 1.0 & 0.956 & 0.621 \\ 1.0 & -0.272 & -0.649 \\ 1.0 & -1.106 & 1.703 \end{bmatrix} \begin{bmatrix} Y \\ I \\ Q \end{bmatrix}$$

5. YCbCr Colour Coordinate.

The YCbCr colour coordinate was developed as part of the ITU-R BT.601 during the establishment of a worldwide digital video component standard. The YCbCr signals are scaled and offset versions of the YIQ format. Y is defined to have a nominal range of 16 to 235; Cb and Cr are defined to have a range of 16 to 240, with zero signal corresponding to level 128. There are several YCbCr formats such as 4:4:4, 4:2:2 and 4:1:1.

IMAGE STORAGE MECHANISMS

Digital imaging systems can store and transmit images. Data is of two types:

1. Human interpretable such as text and number
 2. Diffused data which only machines can interpret.
- If the image files are opened in the normal editor, the information cannot be interpreted meaningfully, as image data is only machine interpretable. Hence image data is called diffused data.
 - To get meaningful information, knowledge of the file format is required.



File format:

File format defines exactly how the image data is stored along with the image metadata in a file.